Classifying Social Media Platforms From Language Patterns

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Objective

Objective and Value Proposition

- Investigate if there is a difference in the language (diversity, style) used across various social media platforms
- Build and compare models that classify which social media platform a post originated from
- Gain insight into the different vocabularies and linguistic styles that arise separately across different platforms

Motivation

- Different platforms have different ‘cultures’
- How are these cultures expressed?
- Wanted to explore: NLP classification, feature engineering, semantic embeddings

Datasets

- Hugging face datasets covering Twitter, Reddit, and Hacker News
  - Twitter:https://huggingface.co/datasets/enryu43/twitter100m_tweets
  - Reddit:https://huggingface.co/datasets/anhchanghoangsg/reddit_pushshift_dataset_cleaned
  - HackerNews:https://huggingface.co/datasets/open-index/hacker-news

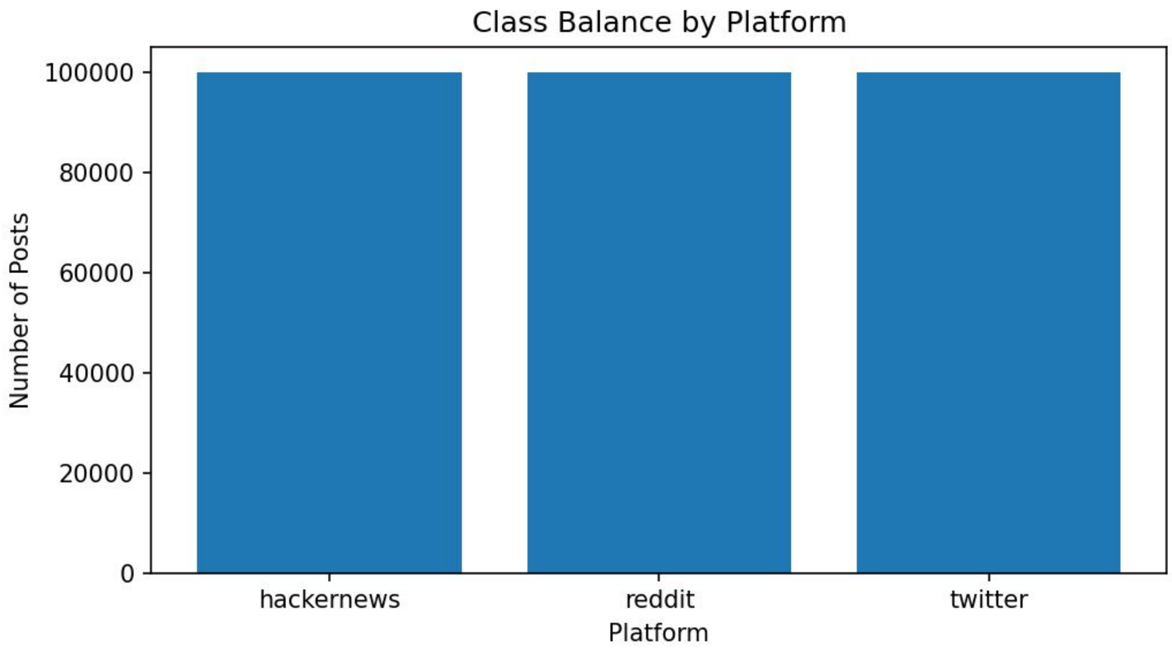
Datasets

- Final Dataset: ~300,000 cleaned posts
- Focus on text-based features
  - Text statistics: post length, word length
  - Punctuation: # questions, explamations, urls, …
  - tf/idf and sentence embeddings

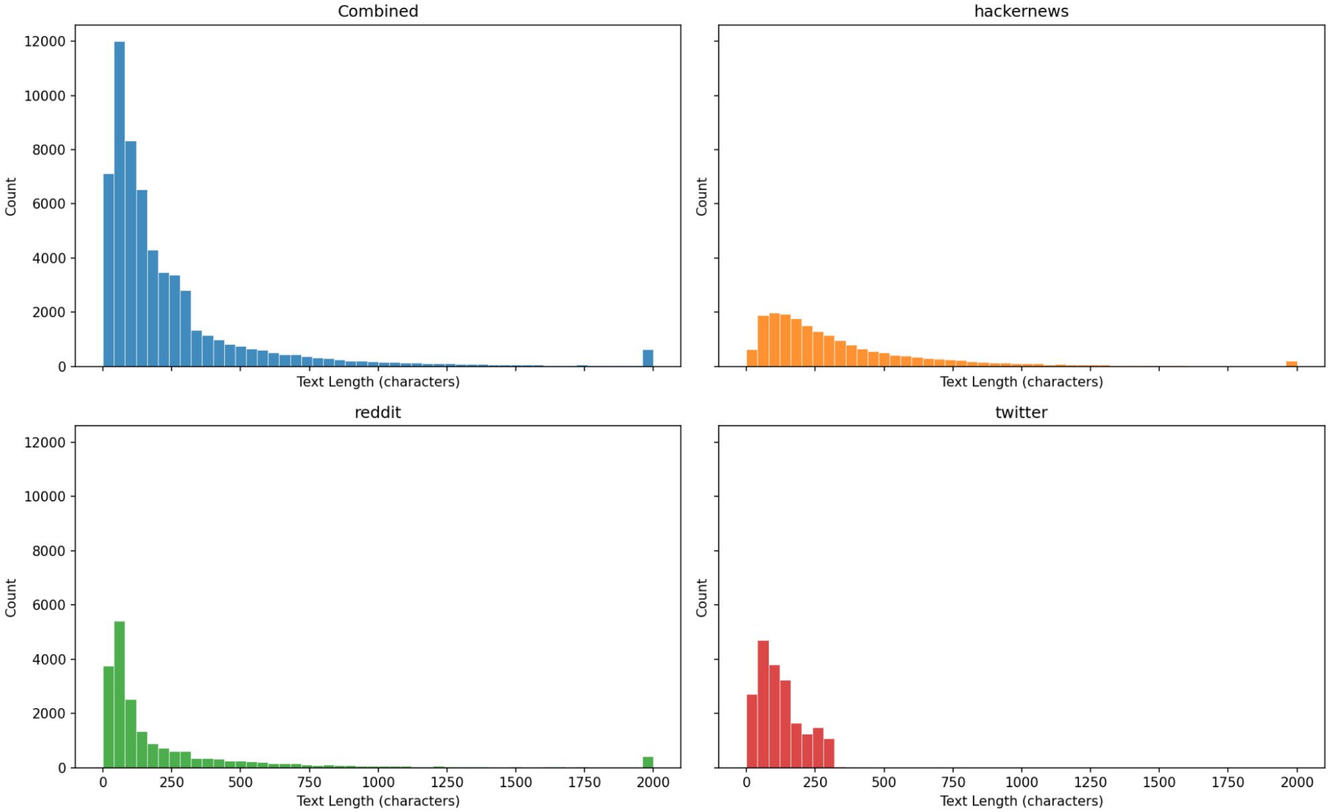
Exploratory Data Analysis

Class Balance by Platform

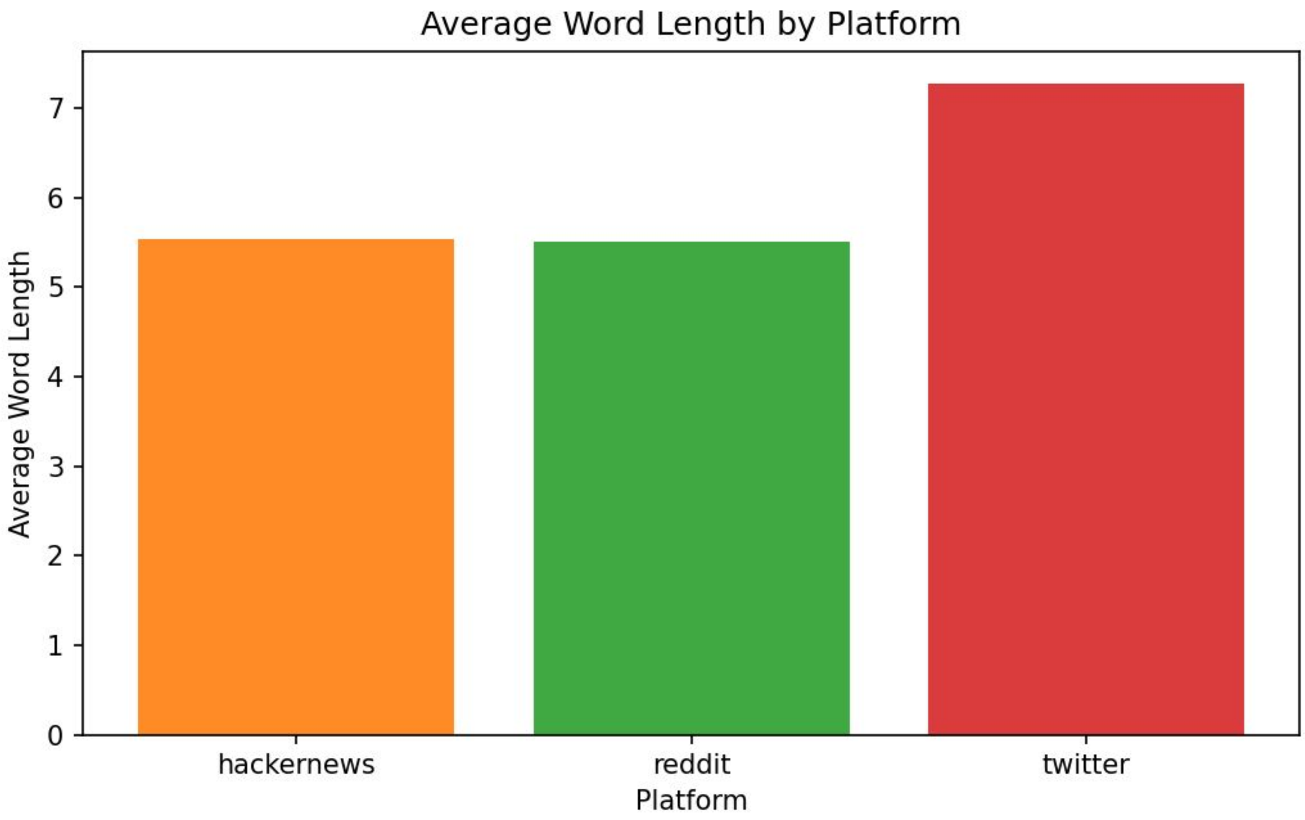
- Make sure data is balanced by sampling from source datasets



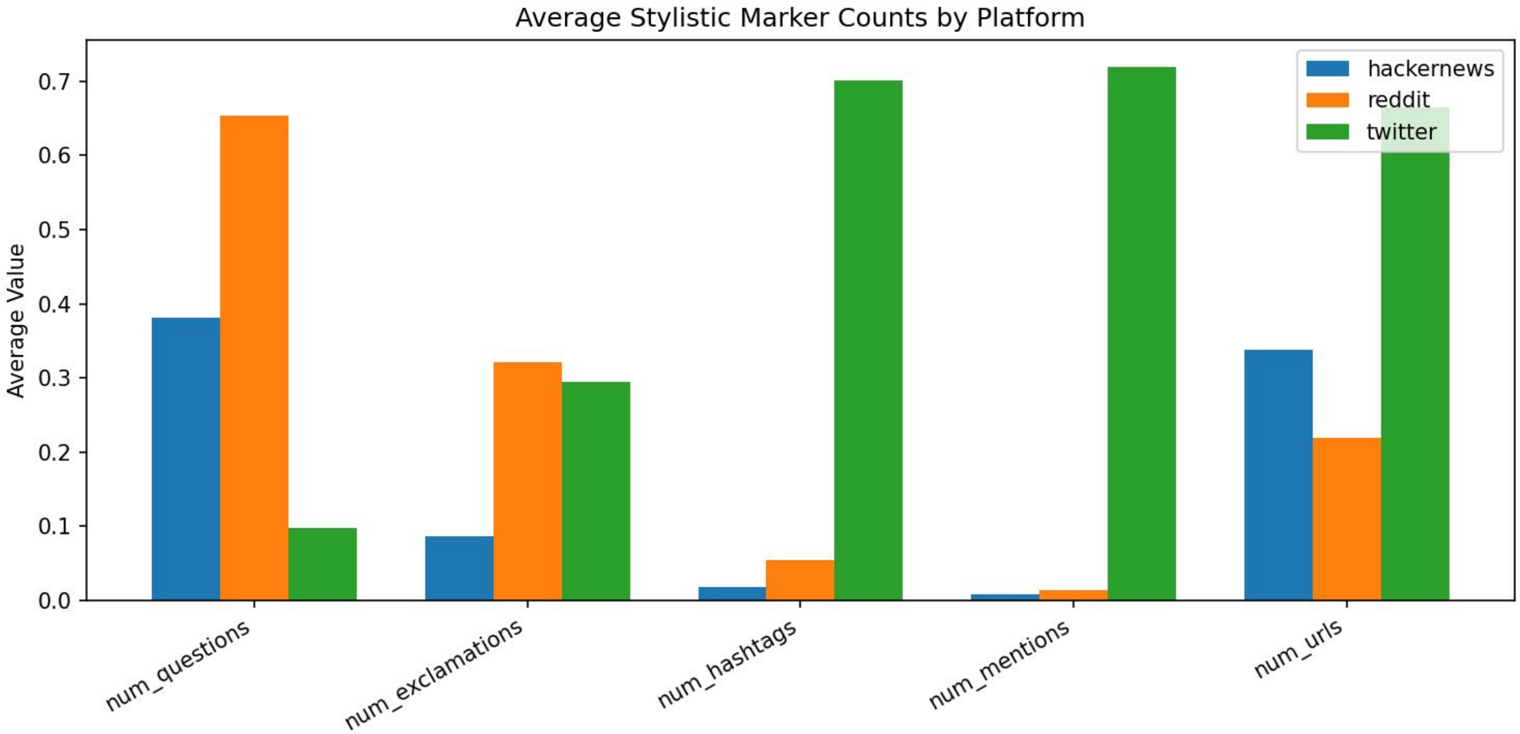
Text Length Distributions (Capped at 2000, Evenly Spaced Bins)



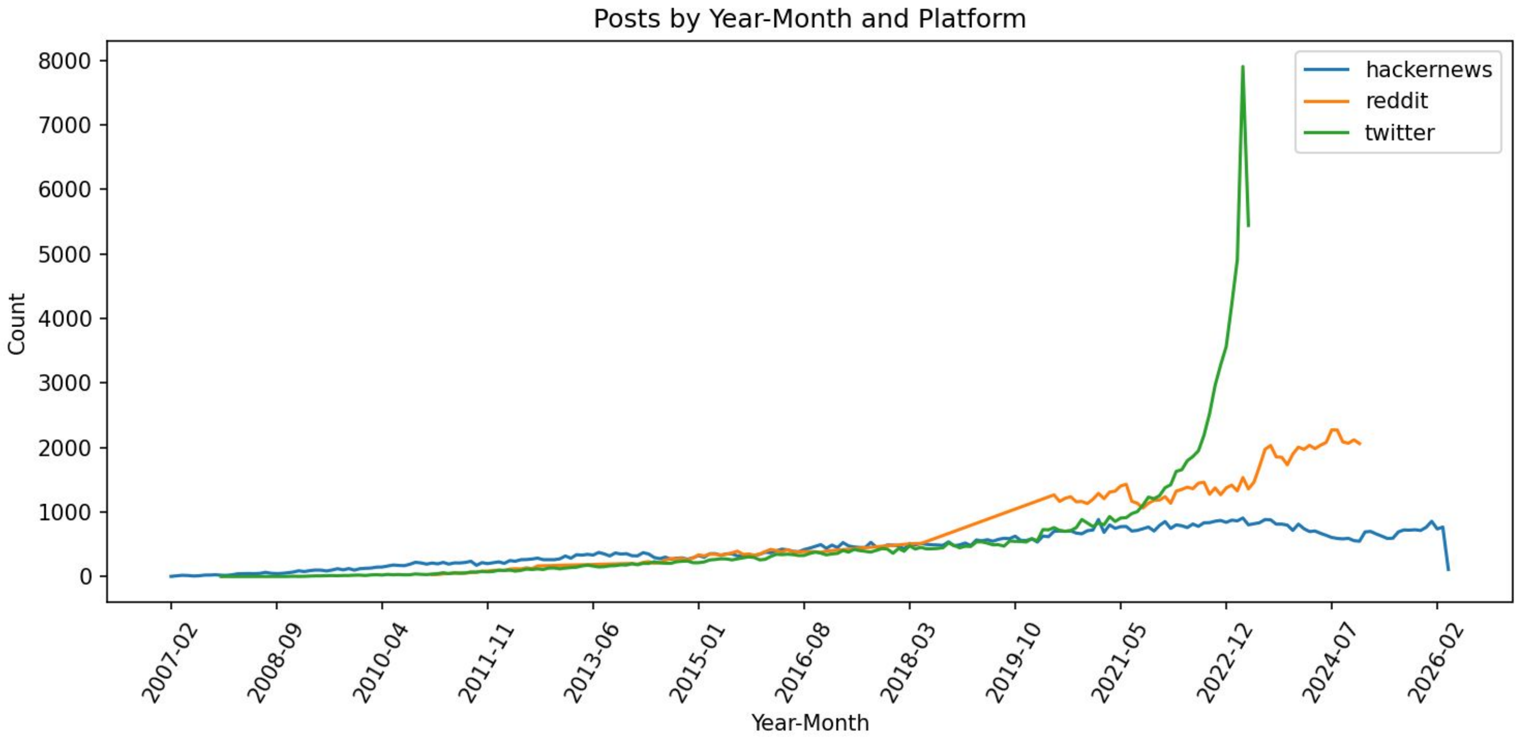
Average Word Length by Platform



Average Stylistic Marker Counts by Platform



Posts by Year-Month and Platform



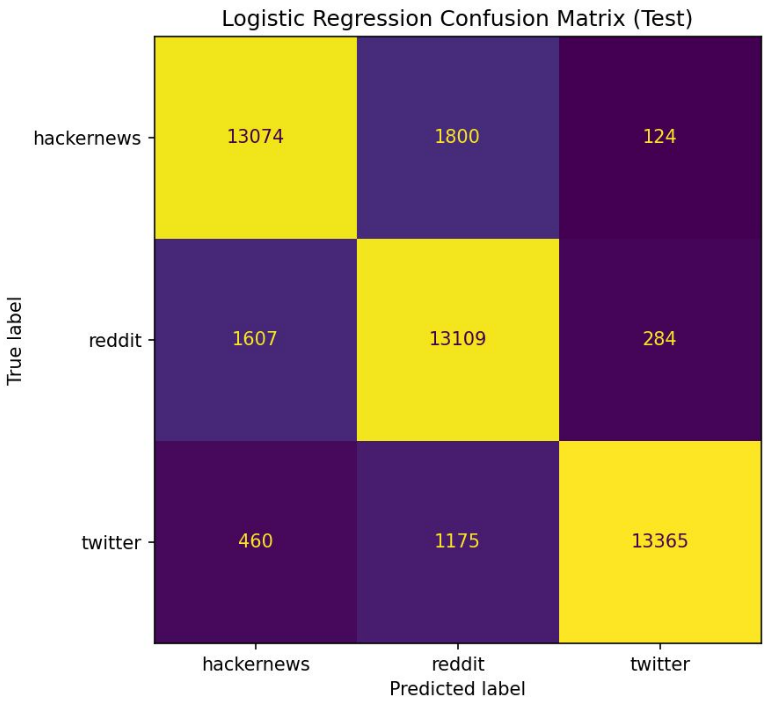
Modeling Results

Models Used

- Logistic Regression, XGBoost, Neural Network
- Features (Text Based)
  - TF-IDF for logistic regression
  - Sentence embeddings (MiniLM) for XG, NN
  - Text length
  - Word count
  - Punctuation counts
  - Capitalization statistics
  - Other stylistic text features (num_questiosn, num_hashtags, num_digits, num_allcaps_words, etc)

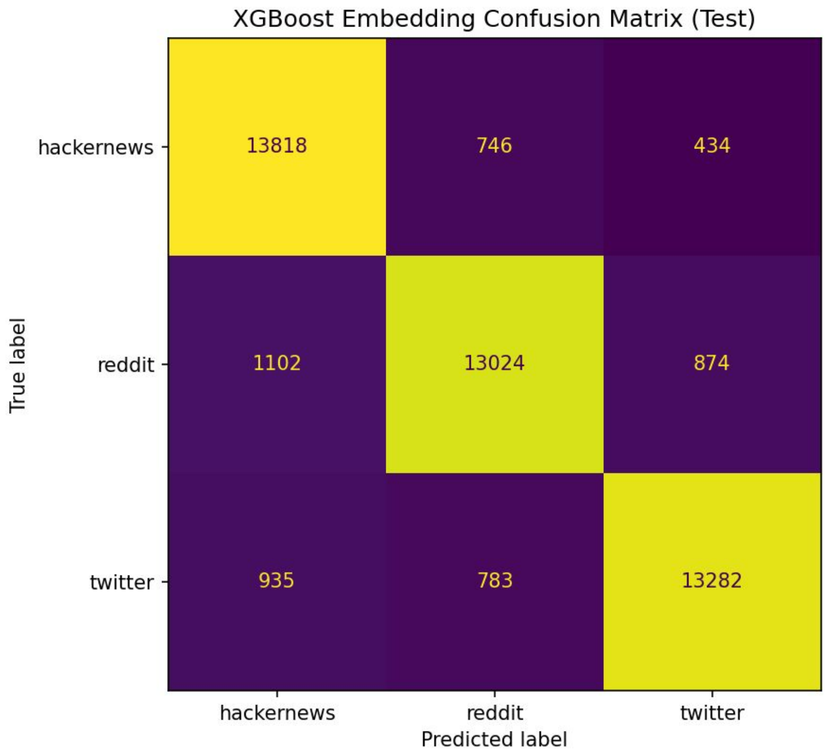
Logistic Regression

- Logistic Regression for a strong, interpretable baseline
- Performance:
  - 87.9% Test Accuracy
  - 0.88 F1-Score



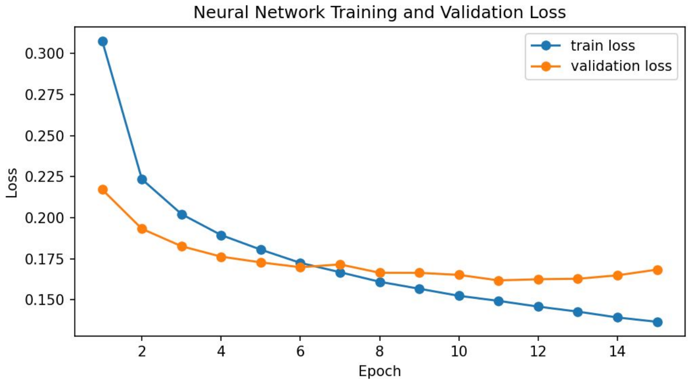
XGBoost

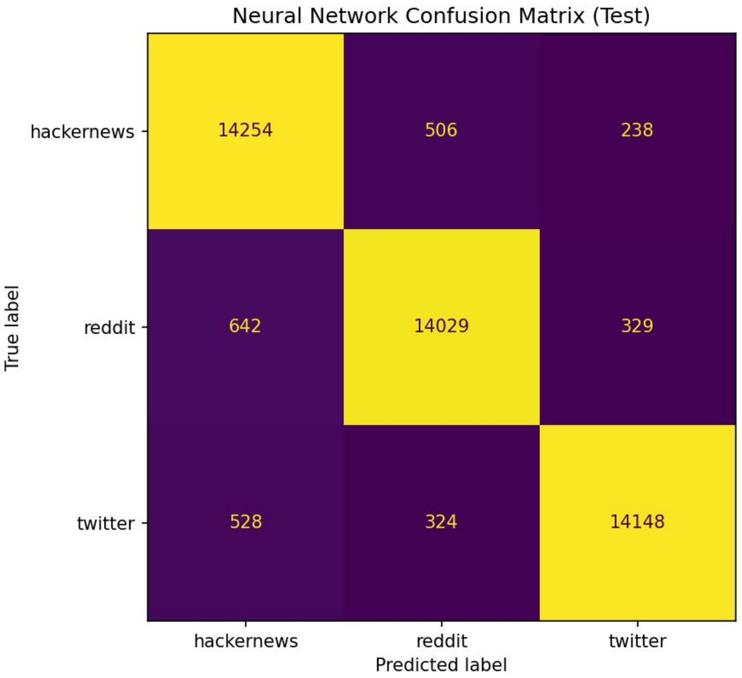
- XGBoost for a stronger but still interpretable model
- Performance:
  - 89.20% Test Accuracy
  - 0.89 F1-Score



Neural Network

- Neural Network maximum performance at the cost of interpretability.
- Performance:
  - 94.3% Test Accuracy
  - 0.94 F1-Score





Implications and Insights

Implications and Insights

- Top Positive Features (Logistic Regression TF/IDF)
- Platform specific language emerges

```
===== Top positive features for reddit =====
telegram       3.014149
looking        2.904652
help           2.568728
overview       2.475501
guys           2.290458
num_words      2.249691
girl           2.189551
poll           2.044982
reddit         2.039501
dm             2.032964
wondering      2.003841
wanna          1.876439
sub            1.876273
reddit com     1.853424
cum            1.843836
Name: reddit, dtype: float64
```

```
===== Top positive features for hackernews =====
com            5.801167
org            4.760560
article        3.214320
hn             3.112926
www            2.962313
probably       2.802742
github         2.788735
point          2.761451
people         2.759692
google         2.611358
doesn          2.523546
yes            2.407572
interesting    2.353612
yeah           2.348904
isn            2.344044
```

```
===== Top positive features for twitter =====
https          14.157866
http            6.082246
num_mentions    2.765396
la              2.271574
00              2.143079
que             1.976868
el              1.526586
today           1.349722
week            1.272876
le              1.221238
tonight         1.178061
1.171919              في
les             1.168812
おはようございます      1.160835
et              1.142039
```

Implications and Insights

- Hacker News: “github”, “articls”, “google”, “point”, “example”
  - Much more technical focused with examples and data
  - Discussion around technology and providing links and articles

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Implications and Insights

- Reddit: “looking”, “help”, “wondering”, “sub”
  - Suggest more conversational, community oriented, advice seeking
  - Reddit specific words like “sub” and “reddit”, informal/explicit language

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Implications and Insights

- Twitter: “https”, “today”, “wek”, “tonight”
  - Highly link driven, event oriented, multi lingual
  - Socially connected, twitter specific style (num_mentions: @)

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Challenges and Future Work

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- Challenges
  - Very large datasets required efficient preprocessing pipelines
  - Balancing interpretability vs performance
  - Presence of unintentional features the models could use to ‘cheat’
- Future Work
  - Add more platforms
  - More advanced transformer-based classifiers
  - Style-transfer or post-generation systems
  - Better explainability tools for embeddings

Thank You

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